Hatim **Kanchwala**

Birth date: 7 February 1995
Address: NIBM Road, Pune 411060 Maharashtra
Phone: +91 897 577 6853
Email: hatim@hatimak.me

Experience

Jan. 2023 – Present
Bengaluru, India

RTL Design Engineer
Qualcomm India Pvt Ltd

- Worked on micro-architecture design and SystemVerilog implementation of Adreno GPU blocks in mobile and compute tier SoC.
- Received appreciation for thorough and high-quality RTL designs and micro-architecture diagrams.
- Executed design cleanup tasks to improve code readability and maintainability across multiple projects.
- Proposed AI initiatives and built automation scripts in Python for auto-genration of SystemVerilog RTL, saving repetitive manual effort across multiple projects and derivatives.
- Analyzed and implemented fixes for timing improvement, power optimization, and wire count reduction.
- Supported verification effort by providing whitebox coverage RTL and formal checks for functional behaviour.
- Supported timing closure effort for max frequency push for compute-tier product lines.
- Collaborated with cross-functional teams including verification, physical design, and architecture to ensure timely delivery of work products.

Oct. 2020 – Dec. 2021
Aachen, Germany

Research Engineer
Forschungszentrum Juelich GmbH

- Developed design with soft-core microprocessors to rapidly prototype control-loop algorithms for FPGA-based real-time simulators.
- Implemented models using High-Level Synthesis designs for RTL co-simulation and real-time simulation.
- Conceptualised heterogenous architecture of control and data-logger soft-cores dedicated to running control algorithms at switching frequency and logging simulation data.
- Implemented digital design based on soft-core MicroBlaze microprocessor from Xilinx on Virtex Ultrascale+ VCU118 board.
- Extended HLS models with memory-mapped `AXI4` register interface and verified on Virtex-7 VC707 FPGA board.
- Developed automation scripts on Linux for code generation, synthesis, and bitstream generation stages.

May 2019 – Sept. 2020
Aachen, Germany

Engineer, Development of FPGA Systems
E.ON Energy

- Integrated Xilinx FPGA boards into VILLAS co-simulation platform by designing an architecture built on top of Aurora 8B/10B serial protocol.
- Engineered automation scripts for design generation and bitstream compilation leading to version-agnostic local toolchain use.
- Engineered Tcl-Makefile system of scripts to automate design generation and bitstream compilation.
- Developed designs in Verilog/VHDL and driver programs in C/C++ for FPGA firmware.

May 2018 – Nov. 2018
DRDO

Senior Research Engineer
Naval Physical & Oceanographic Laboratory

- Implemented advanced tracking filters in MATLAB, like Distributed Extended Kalman Filter, Shifted Rayleigh Filter, and Sequential Monte Carlo methods, for the Bearings-only Tracking problem.
- Simulated performance of modern filters on real field manoeuvre data from Indian Navy, and prepared comparative study.
- Concluded Shifted Rayleigh Filter outperforms other filters in terms of computational complexity and tracking accuracy.

Aug. 2017 – May 2018
IIT Patna, India

Research Engineer
Control and Instrumentation Lab

- Designed and implemented parallel architecture for "family of Sigma-Point Kalman filtering algorithms" on FPGA.
- Conceptualised novel parallel routine for Cholesky matrix decomposition; improvement from $O(N^3)$ to $O(N)$ time complexity. Customised and implemented a matrix inversion routine to compute covariance inverses in $O(N)$ time complexity.
- Optimised resource usage of Cholesky decomposition architecture for double utilisation at same processor count.
- Implemented parallel designs in Verilog HDL using Vivado and open-source IPs on Zynq-7000 ZC702 and Digilent Nexys4 DDR FPGA boards.

Education

Apr. 2019 – Oct. 2022
Aachen, Germany

M. Sc. Electrical Engineering, Information Technology and Computer Engineering
RWTH Aachen University
Final Grade 2,1 (German) = 8 / 10 (Indian)
Thesis: "Field-Programmable Gate Array based Real-Time Control and Simulation"

July 2014 – May 2018
Bihta (Patna), India

B. Tech. Electrical Engineering
Indian Institute of Technology Patna
Final Grade 7.32 / 10
Thesis: "Hardware Architecture of a Family of Sigma-Point Kalman Filters for Bayesian Estimation"

- Nominated for Best B. Tech. Thesis award from Dept of Electrical Engineering.
- One of only two students to receive 10 / 10 grade.

2001 – 2011
Aurangabad, India

Class X Central Board of Secondary Education
Nath Valley School
Final Grade 10 / 10

Skills

Programming SystemVerilog, C/C++, Python, JavaScript, Shell/Tcl, Claude Code

Software Verdi, Xilinx Vivado & HLS, MATLAB, Simulink, `git`/GitHub, RSCAD, NI LabVIEW, GNU/Linux, `gdb`, Verilator, `yosys`, LaTeX, `gnuplot`

Hardware Xilinx Virtex & Zynq SoC, Digilent Nexys4 DDR, Lattice FPGA, Embedded Microcontrollers

Languages

English	*Listening* C2	*Reading* C2	*Speaking* C2	*Writing* C2	*TOEFL* 114 / 120
Deutsch	*Hören* B1	*Lesen* B2	*Sprechen* B1	*Schreiben* B1	
Hindi	native				

Achievements

2019 – 2021
Aachen, Germany

Awarded monthly stipend for research project work
Institute for Automation of Complex Power Systems, RWTH Aachen

2017 – 2018
IIT Patna

Nominated for Best B.Tech. Thesis award
Dept of Electrical Engineering

2017
Hebden Bridge, UK

Presented "EDSAC Museum on FPGA" at digital design conference with full sponsorship
ORConf 2017

2014
Pune, India

Secured All India Rank 4997 (99.6156th percentile) in JEE Advanced
All India Joint Entrance Exam

2010
New Delhi, India

Secured rank 25 and received gold medal in the final round
9th National Cyber Olympiad

2008
Aurangabad, India

Secured rank 156 in the final round
7th National Cyber Olympiad